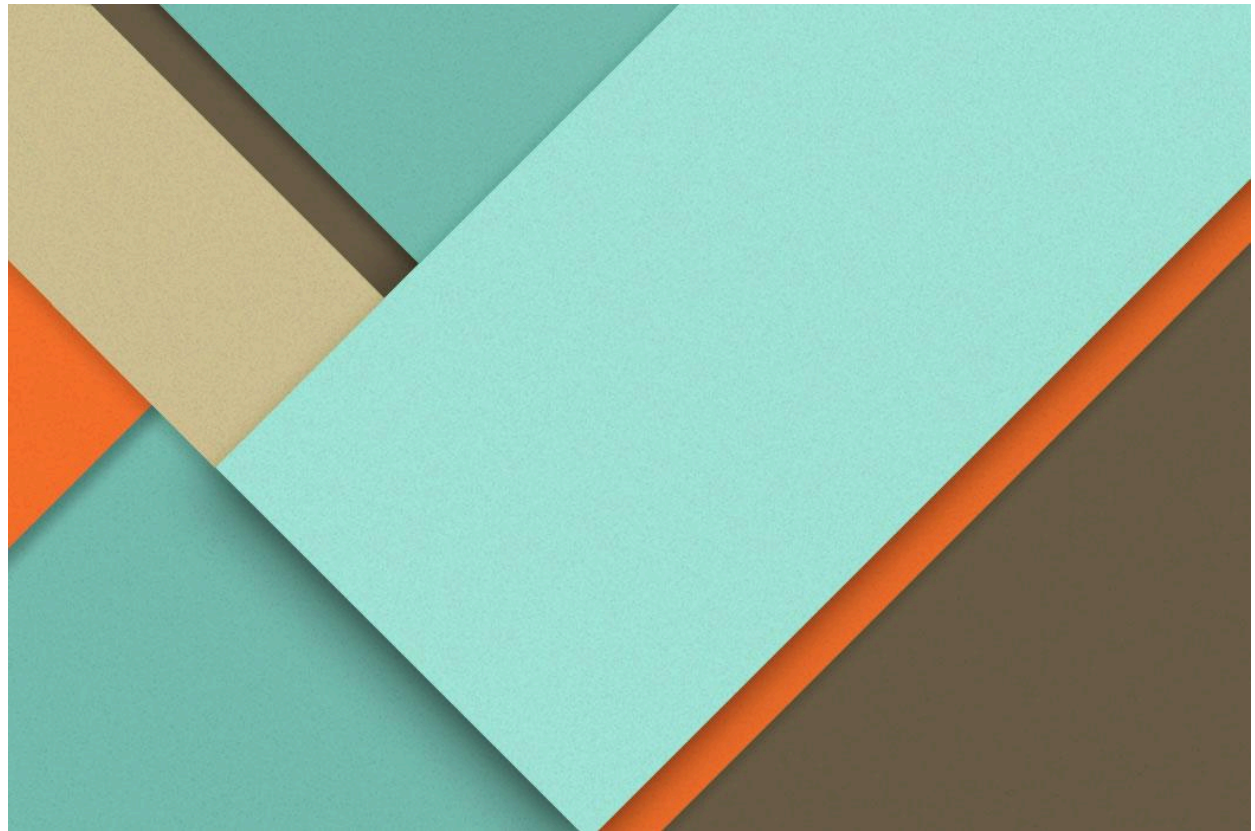




Solving Inventory Deficiencies using SQL

Executive Summary

Eshan Sugeesh E R

Engineering Physics PART - IV

IIT (BHU) Varanasi

Organisation: Consulting & Analytics Club, IIT Guwahati

Background

Urban Retail Co., a growing mid-sized retail chain with over 5,000 SKUs across online and offline stores, is experiencing inventory management challenges, including:

- Frequent stockouts of high-demand products
- Overstocking of slow-moving items, tying up working capital
- Lack of real-time SKU-level visibility
- Absence of data-driven decision-making processes

This project aimed to leverage SQL analytics and BI dashboards to monitor, diagnose, and optimise inventory flow using the company's sales and supply chain data.

Key Insights

Missed Sales:

Products like P0066 and P0126 in the South and West regions consistently failed to meet forecasted demand, especially during seasonal spikes and promotional periods.

Estimated loss of potential revenue during high-demand windows due to inadequate replenishment planning and slow reaction time.

Inventory Turnover:

The clothing category had the highest turnover.

Store S001 in the East and Store S003 in the West had the highest turnover among stores.

Reorder Point Logic:

This logic simulates a real-world buffer stock system where businesses want to ensure at least 3 days' worth of average sales are always available to avoid stockouts.

Products like P0057 are hitting the highest average daily sales, indicating the need for proactive replenishment policies.

Overstocking Logic:

An analysis of inventory levels versus average demand was carried out to detect overstock beyond the 3-day buffer threshold. Product P0125 have inventory exceeding demand projections by over 126 units, and many others with 87 overstock units, tying up working capital unnecessarily

Recommendations

I. Reduce Stockouts of Fast-Moving Products

- Implement automated reorder alerts based on historical average sales.
- Maintain minimum stock thresholds (e.g., 3× avg daily sales) for fast-movers.
- Integrate demand forecasting into replenishment plans.

II. Address Overstocking of Slow-Moving Products

- Use the overstocking SQL logic to identify slow-movers.
- Apply dynamic pricing or discounts to clear excess stock.
- Reassess reorder cycles for these products to reduce future accumulation.

III. Optimise Inventory Turnover

- Prioritise the procurement of high-turnover SKUs and limit stock for low-performing ones.
- Build category-wise and store-wise turnover benchmarks.

IV. Improve Demand Forecast Accuracy

- Refine forecasting logic using time series methods (e.g., rolling averages, exponential smoothing).
- Factor in seasonality and weather more aggressively.
- Retrain forecast models quarterly using fresh sales data.

V. Enhance Visibility Across Regions and Stores

- Identify and replicate practices from top-performing stores.
- Reallocate stock based on regional demand insights.

VI. Build a Robust Data Infrastructure

- Normalise raw data into a fact-dimension structure.
- Use indexes and optimised joins for query performance.
- Ensure consistency in SKUs, stores, dates, and categories.

By adopting these recommendations, Urban Retail Co. can reduce missed revenue, unlock working capital, and enhance customer satisfaction through smarter, data-driven inventory planning.